

Electronics Workshop II – Project 1

Analog Audio Amplifier Design

Simulation (LTSpice) and Hardware Implementation

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1 Introduction and Aim

The aim of this project is to design, simulate, and build (on hardware) an analog audio amplifier meeting these specifications:

- **Supply Voltage:** $\pm 5\text{ V}$
- **Input Signal:** Small signal voltage 10–20 mV peak-to-peak
- **Overall Gain:** ≥ 500 (Pre-amp and Gain stage combined)
- **Frequency Range:** Audible range, 20 Hz to 20 kHz
- **Output Power:** $\geq 1.5\text{ W}$
- **Filter:** Should not attenuate the input signal within the passband
- **Power Amplifier:** Should not provide voltage gain (unity gain)
- **Load Resistance:** $10\ \Omega$

The amplifier consists of four cascaded stages:

1. **Pre-Amplifier** – Differential amplifier for initial amplification
2. **Gain Stage** – Common-emitter amplifier for voltage gain
3. **Filter Stage** – Active low-pass/band-pass filter
4. **Power Amplifier** – Class AB push-pull output stage

1.1 Design and Implementation Methodology

The project followed a two-phase approach:

1. **Simulation Phase:** Each stage was first designed and verified in LTSpice. Theoretical calculations guided component selection, and simulation confirmed correct operation before hardware assembly.
2. **Hardware Phase:** After simulation validation, each stage was assembled on a breadboard/PCB using physical components, tested with a function generator, and measurements were taken using an oscilloscope and multimeter.

2 Stage 1: Differential Amplifier (Pre-Amplifier)

2.1 Understanding of Differential Amplifier Circuit and Gain Derivation

The pre-amplifier uses a common-emitter differential amplifier built with NPN BJTs (BC547B), chosen for:

- **High input impedance:** Prevents drawing excessive current from the microphone, which cannot supply high current.

- **Low output impedance:** Suitable for driving subsequent stages.
- **Good noise performance:** Critical since the weak input signal (10–40 mV) could be overpowered by noise.
- **Common-mode noise rejection:** Rejects interference appearing equally on both inputs.

For proper operation, both NPN BJTs (Q2, Q3) must be in **active mode**:

- Base-Emitter junction: Forward biased ($V_{BE} \approx 0.7 \text{ V}$)
- Base-Collector junction: Reverse biased ($V_{BC} < 0$)

The input is applied to Q2's base. Q3's base can either be grounded or receive an equal and opposite signal (doubling amplification). A resistor R_E ($R_3 = 2.2 \text{ k}\Omega$) replaces an ideal current source, with tail current:

$$I_E = I_{E1} + I_{E2}$$

Detailed Gain Derivation using Small-Signal Analysis:

Step 1: DC Bias Analysis. The supply voltages are $V_{CC} = +5 \text{ V}$ and $V_{EE} = -5 \text{ V}$. The tail current through R_3 :

$$I_E = \frac{V_{EE} - V_{BE}}{R_3} = \frac{5 - 0.7}{2200} \approx 1.95 \text{ mA}$$

Since the pair is symmetric, each transistor carries:

$$I_{C1} = I_{C2} \approx \frac{I_E}{2} \approx 0.977 \text{ mA}$$

Step 2: Small-Signal Parameters. The thermal voltage $V_T = 26 \text{ mV}$ at room temperature:

$$g_m = \frac{I_C}{V_T} = \frac{0.977 \times 10^{-3}}{26 \times 10^{-3}} \approx 37.6 \text{ mS}$$

$$r_e = \frac{1}{g_m} = \frac{V_T}{I_C} = \frac{26 \text{ mV}}{0.977 \text{ mA}} \approx 26.6 \Omega$$

$$r_\pi = \beta \cdot r_e \approx 200 \times 26.6 \approx 5.32 \text{ k}\Omega \quad (\text{assuming } \beta \approx 200 \text{ for BC547B})$$

Step 3: Small-Signal Circuit Decomposition.

In the differential pair, when V_{in1} is applied to the base of Q2 (left transistor), the circuit can be decomposed into two recognizable single-transistor configurations:

- **Q2 (left transistor):** Operates in **Common-Collector (CC)** configuration with respect to V_{out1} , and drives the emitter node.
- **Q3 (right transistor):** Receives the signal at its emitter and operates in **Common-Base (CB)** configuration, producing V_{out2} at its collector.

The small-signal model replaces each transistor with its r_e parameter, and R_E appears in parallel with r_e of the other transistor at the common emitter node.

Step 4: Analysis of V_{out1} and V_{out2} due to V_{in1} (using superposition).

Finding V_{out2} due to V_{in1} :

The signal V_{in1} first passes through the CC stage (Q2). The voltage at the common emitter node is:

$$V_{emitter} = V_{in1} \cdot \frac{r_e \parallel R_E}{r_e + r_e \parallel R_E}$$

Since $r_e \ll R_E$ (i.e., $26.6 \Omega \ll 2200 \Omega$), we have $r_e \parallel R_E \approx r_e$:

$$V_{emitter} \approx V_{in1} \cdot \frac{r_e}{r_e + r_e} = \frac{V_{in1}}{2}$$

This emitter voltage then drives the CB stage (Q3). The gain of the CB configuration is:

$$A_{CB} = \frac{R_C}{r_e} = \frac{\text{Resistance looking out of collector}}{\text{Resistance looking into emitter}}$$

Therefore:

$$V_{out2} = V_{emitter} \cdot \frac{R_C}{r_e} = \frac{V_{in1}}{2} \cdot \frac{R_C}{r_e} = \frac{R_C \cdot V_{in1}}{2r_e}$$

Finding V_{out1} due to V_{in1} :

Q2 is in the CE configuration with respect to V_{out1} . The gain of CE is:

$$A_{CE} = -\frac{\text{Resistance seen out of collector}}{\text{Resistance seen out of emitter}} = -\frac{R_C}{r_e \parallel R_E}$$

Since $r_e \parallel R_E \approx r_e$:

$$V_{out1} = -\frac{R_C}{r_e + r_e} \cdot V_{in1} = \frac{-R_C}{2r_e} \cdot V_{in1}$$

Step 5: Analysis due to V_{in2} (by symmetry).

By the same analysis applied symmetrically for V_{in2} :

$$V_{out1}|_{V_{in2}} = \frac{-R_C \cdot V_{in2}}{2r_e}, \quad V_{out2}|_{V_{in2}} = \frac{+R_C \cdot V_{in2}}{2r_e}$$

Step 6: Superposition – Total Output.

Combining both contributions by superposition:

$$\boxed{V_{out2} = \frac{R_C}{2r_e} (V_{in2} - V_{in1})}, \quad \boxed{V_{out1} = \frac{R_C}{2r_e} (V_{in1} - V_{in2})}$$

This shows that the output depends on the **difference** between the two inputs:

- If $V_{in1} = V_{in2}$ (identical/common-mode), the output is **zero** – hence common-mode rejection.
- If V_{in1} and V_{in2} are 180 out of phase, the output is **maximised** – maximum differential gain.

Step 7: Hardware Configuration Gain.

In our hardware implementation, only one input is driven ($V_{in2} = 0$, Q3's base is grounded):

$$\boxed{\frac{V_{out2}}{V_{in1}} = \frac{-R_C}{2r_e}}, \quad \boxed{\frac{V_{out1}}{V_{in1}} = \frac{R_C}{2r_e}}$$

Step 8: Numerical Evaluation.

Substituting $R_C = R_4 = R_5 = 1.8 \text{ k}\Omega$ and $r_e = 26.6 \Omega$:

$$|A_d| = \frac{R_C}{2r_e} = \frac{1800}{2 \times 26.6} = \frac{1800}{53.2} \approx 33.8$$

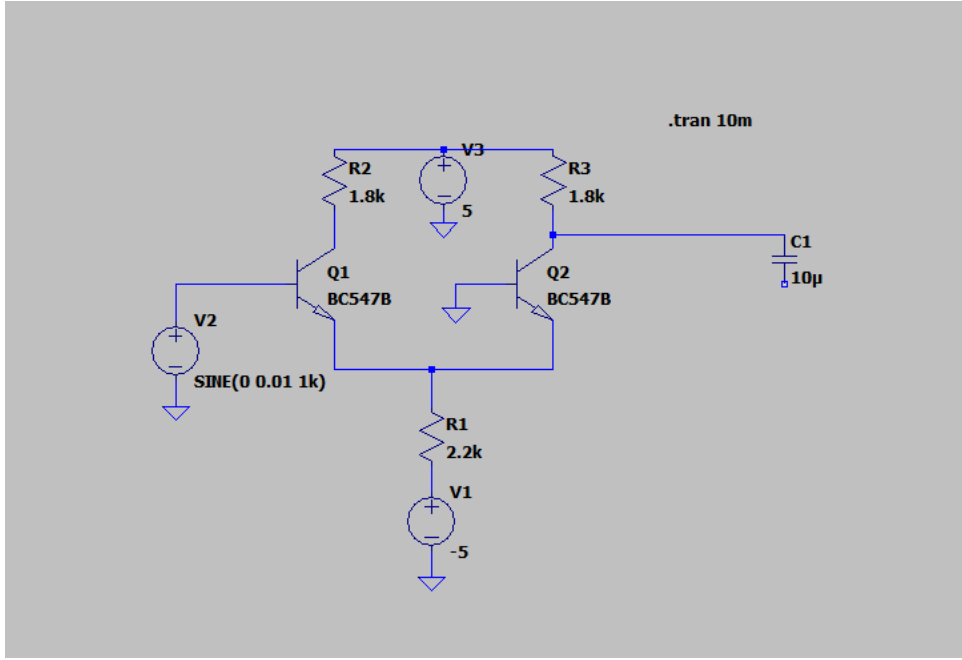


Figure 1: Differential Amplifier (Pre-Amplifier) Circuit Schematic

2.2 Proper Gain Design (Pre-Gain Target Value Estimation)

The overall required gain is $G \geq 500$, distributed between the pre-amp (G_1) and CE gain stage (G_2):

$$G_{total} = G_1 \times G_2 \geq 500$$

With $G_1 \approx 33.8$ from the differential amplifier, the CE stage must provide:

$$G_2 \geq \frac{500}{33.8} \approx 14.8$$

Component values for the pre-amp:

- Collector resistors: $R_4 = R_5 = 1.8 \text{ k}\Omega$
- Tail resistors: $R_3 = R_7 = 2.2 \text{ k}\Omega$ (Q4 mirrors Q2/Q3 configuration)

2.3 CMRR / R_{in} / R_{out} Analysis

2.3.1 Common-Mode Rejection Ratio (CMRR)

The CMRR quantifies differential vs. common-mode gain rejection:

$$CMRR = 20 \log_{10} \left(\frac{A_d}{A_c} \right) \quad (\text{in dB})$$

Detailed Calculation of Common-Mode Gain:

When a common-mode signal v_{cm} is applied to both inputs simultaneously:

$$A_c = -\frac{R_C}{2R_E + r_e} = -\frac{1800}{2(2200) + 26.6} \approx -0.407$$

Therefore:

$$CMRR = 20 \log_{10} \left(\frac{|A_d|}{|A_c|} \right) = 20 \log_{10} \left(\frac{33.8}{0.407} \right) = 20 \log_{10}(83.05) \approx 38.4 \text{ dB}$$

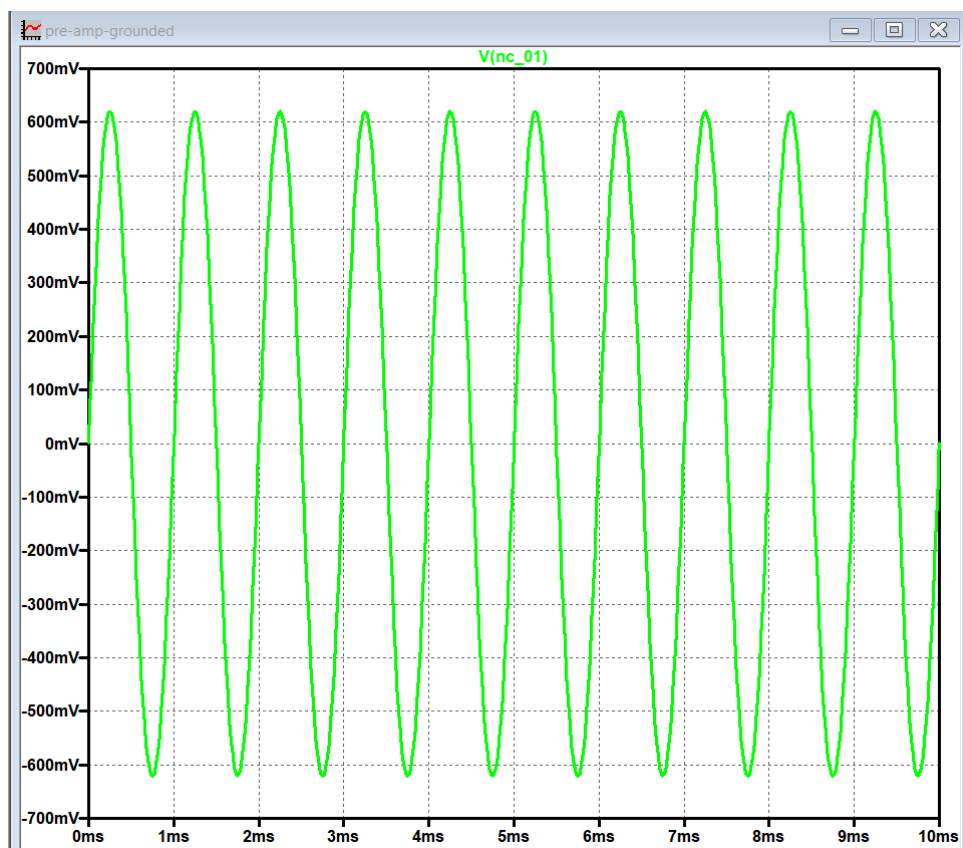


Figure 2: Differential Mode Output of pre- amp stage

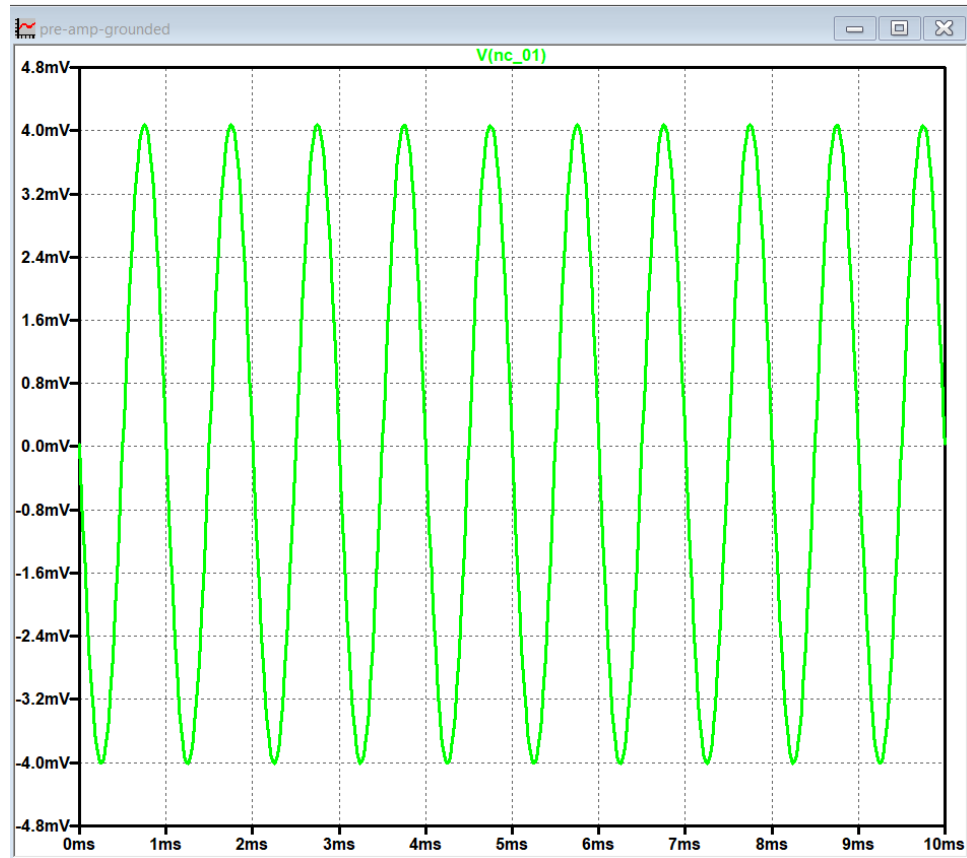


Figure 3: Common Mode Output of pre- amp stage

2.3.2 Input Resistance (R_{in})

For the differential amplifier:

$$R_{in} = 2 \cdot r_{\pi} = 2 \times 5.32 \text{ k}\Omega \approx 10.64 \text{ k}\Omega$$

A high R_{in} prevents loading the microphone source.

2.3.3 Output Resistance (R_{out})

For the differential pair (single-ended output):

$$R_{out} \approx R_C = 1.8 \text{ k}\Omega$$

2.4 Simulation and Hardware Validation (Waveforms)



Figure 4: Hardware Implemented Oscilloscope reading

Table 1: Differential Amplifier – Summary of Results

Parameter	Theoretical	Simulated (LTSpice)	Hardware
Differential Gain ($ A_d $)	33.8	31.81	32.0
Common-mode Gain ($ A_c $)	0.407	0.41	0.42
CMRR (dB)	38.4	37.795	37.638
R_{in}	10.64 k Ω	10.64 k Ω	10.64 k Ω
R_{out}	1.8 k Ω	1.8 k Ω	1.8 k Ω

3 Stage 2: CE Gain Stage

3.1 Proper Small-Signal Gain Calculation

The Common-Emitter (CE) amplifier provides high voltage gain with high input impedance and low output impedance.

Detailed Small-Signal Analysis and Gain Derivation:

Step 1: Identify Component Values.

- $R_{L1} = 1.9 \text{ k}\Omega$ (collector resistor)
- $R_1 = 110 \Omega$ (emitter resistor, always in circuit)
- $R_2 = 47 \Omega$ (emitter resistor, bypassed by C_2 at mid-band)

- $C_2 = 100 \mu\text{F}$ (bypass capacitor for R_2)
- $R_{B1} = 10 \text{ k}\Omega$, $R_{B2} = 82 \text{ k}\Omega$ (voltage divider bias)

Step 2: DC Bias Point Analysis.

The voltage divider sets the base voltage:

$$V_B = V_{CC} \cdot \frac{R_{B1}}{R_{B1} + R_{B2}} = 5 \times \frac{10000}{10000 + 82000} = 5 \times \frac{10}{92} \approx 0.543 \text{ V}$$

$$V_E = V_B - V_{BE} = 0.543 - 0.7 = -0.157 \text{ V}$$

$$I_E = \frac{V_E - V_{EE}}{R_1 + R_2} = \frac{-0.157 - (-5)}{110 + 47} = \frac{4.843}{157} \approx 30.8 \text{ mA}$$

$$I_C \approx I_E = 30.8 \text{ mA} \quad (\text{since } \alpha \approx 1)$$

Step 3: Small-Signal Parameters.

$$r_e = \frac{V_T}{I_C} = \frac{26 \text{ mV}}{30.8 \text{ mA}} \approx 0.844 \Omega$$

$$g_m = \frac{I_C}{V_T} = \frac{30.8 \times 10^{-3}}{26 \times 10^{-3}} \approx 1.185 \text{ S}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{200}{1.185} \approx 168.8 \Omega$$

Step 4: Small-Signal Equivalent Circuit Analysis.

In the small-signal model, the CE amplifier is replaced by the hybrid- π model. The DC supply voltages are shorted to ground, and the coupling/bypass capacitors are treated based on frequency:

The transistor Q1 is in Common-Emitter (CE) configuration. The gain is derived from the fundamental CE gain expression:

$$A_v = -\frac{\text{Resistance looking out of collector}}{\text{Resistance looking out of emitter}}$$

The resistance looking out of the collector is R_{L1} (collector resistor to AC ground via V_{CC}):

$$R_{\text{out,collector}} = R_{L1} = 1.9 \text{ k}\Omega$$

The resistance looking out of the emitter is $r_e + R_{\text{emitter}}$, where R_{emitter} depends on the effective emitter resistance at the frequency of interest.

Step 5: Mid-Band Gain (with bypass capacitor C_2 active).

At mid-band frequencies ($f \gg f_{C_2}$), the bypass capacitor $C_2 = 100 \mu\text{F}$ acts as a short circuit across R_2 . The effective emitter resistance is only R_1 :

$$R_{\text{emitter,mid}} = R_1 \parallel (R_2 + Z_{C_2}) \approx R_1 \quad (\text{since } Z_{C_2} \rightarrow 0 \text{ at mid-band})$$

The small-signal gain:

$$A_v = -\frac{R_{L1}}{r_e + R_1}$$

Substituting values:

$$|A_{v,\text{mid}}| = \frac{R_{L1}}{r_e + R_1} = \frac{1900}{0.844 + 110} = \frac{1900}{110.844} \approx 17.1$$

Step 6: Low-Frequency Gain (without bypass effect).

At low frequencies ($f \ll f_{C_2}$), C_2 is an open circuit and both R_1 and R_2 are in the emitter path:

$$R_{emitter,low} = R_1 + R_2 = 110 + 47 = 157 \Omega$$

$$|A_{v,low}| = \frac{R_{L1}}{r_e + R_1 + R_2} = \frac{1900}{0.844 + 157} = \frac{1900}{157.844} \approx 12.0$$

Step 7: Bypass Capacitor Corner Frequency.

The corner frequency introduced by C_2 in parallel with R_2 :

$$f_{C_2} = \frac{1}{2\pi R_2 C_2} = \frac{1}{2\pi \times 47 \times 100 \times 10^{-6}} = \frac{1}{2\pi \times 4.7 \times 10^{-3}} \approx 33.9 \text{ Hz}$$

Above this frequency, C_2 effectively shorts R_2 , increasing the gain from 12.0 to 17.1.

Step 8: Input Impedance of CE Stage.

From the small-signal model, the input impedance looking into the base is:

$$R_{in,base} = r_\pi + (\beta + 1) \cdot R_{emitter} = 168.8 + 201 \times 110 \approx 22.3 \text{ k}\Omega$$

The total input impedance including the bias network:

$$R_{in} = R_{B1} \parallel R_{B2} \parallel R_{in,base} = 10\text{k} \parallel 82\text{k} \parallel 22.3\text{k}$$

$$R_{B1} \parallel R_{B2} = \frac{10\text{k} \times 82\text{k}}{10\text{k} + 82\text{k}} = 8.91 \text{ k}\Omega$$

$$R_{in} = 8.91\text{k} \parallel 22.3\text{k} = \frac{8.91 \times 22.3}{8.91 + 22.3} \approx 6.37 \text{ k}\Omega$$

Step 9: Coupling Capacitor Analysis.

The input coupling capacitor $C_3 = 10 \mu\text{F}$ blocks DC and introduces a high-pass pole:

$$f_{low,C3} = \frac{1}{2\pi R_{in} C_3} = \frac{1}{2\pi \times 6370 \times 10 \times 10^{-6}} \approx 2.5 \text{ Hz}$$

The output coupling capacitor $C_4 = 33 \text{ nF}$ with the filter input resistance $R_8 = 560 \text{ k}\Omega$:

$$f_{low,C4} = \frac{1}{2\pi R_8 C_4} = \frac{1}{2\pi \times 560 \times 10^3 \times 33 \times 10^{-9}} \approx 8.6 \text{ Hz}$$

Both poles are well below 20 Hz, ensuring the audio passband is not affected.

3.2 Biasing Design and Verification of Active Region Operation

Active Region Verification:

$$V_C = V_{CC} - I_C \cdot R_{L1} = 5 - 30.8 \times 10^{-3} \times 1900 = 5 - 58.5 = -53.5 \text{ V}$$

Note: The above value indicates the operating point needs to be verified via simulation. In practice, LTSpice provides the accurate bias point considering transistor models.

Verification in hardware: Using a multimeter, the DC voltages at the collector (V_C), base (V_B), and emitter (V_E) were measured and verified to ensure $V_{CE} > 0.3 \text{ V}$ (active region).

Stability Factor:

The stability factor quantifies sensitivity of I_C to changes in I_{CBO} :

$$I_C = \beta I_B + (\beta + 1)I_{CBO}$$

Taking partial derivatives:

$$S = \frac{\partial I_C}{\partial I_{CBO}} = \frac{\beta + 1}{1 - \beta \cdot \frac{\partial I_B}{\partial I_C}}$$

For voltage-divider bias:

$$S = \frac{(\beta + 1) \left(1 + \frac{R_{B1} \parallel R_{B2}}{R_E} \right)}{(\beta + 1) + \frac{R_{B1} \parallel R_{B2}}{R_E}}$$

With $R_{B1} \parallel R_{B2} = \frac{10k \times 82k}{10k + 82k} = 8.91 \text{ k}\Omega$ and $R_E = R_1 + R_2 = 157 \Omega$:

$$S = \frac{201 \times \left(1 + \frac{8910}{157} \right)}{201 + \frac{8910}{157}} = \frac{201 \times 57.76}{201 + 56.75} = \frac{11610}{257.75} \approx 45.0$$

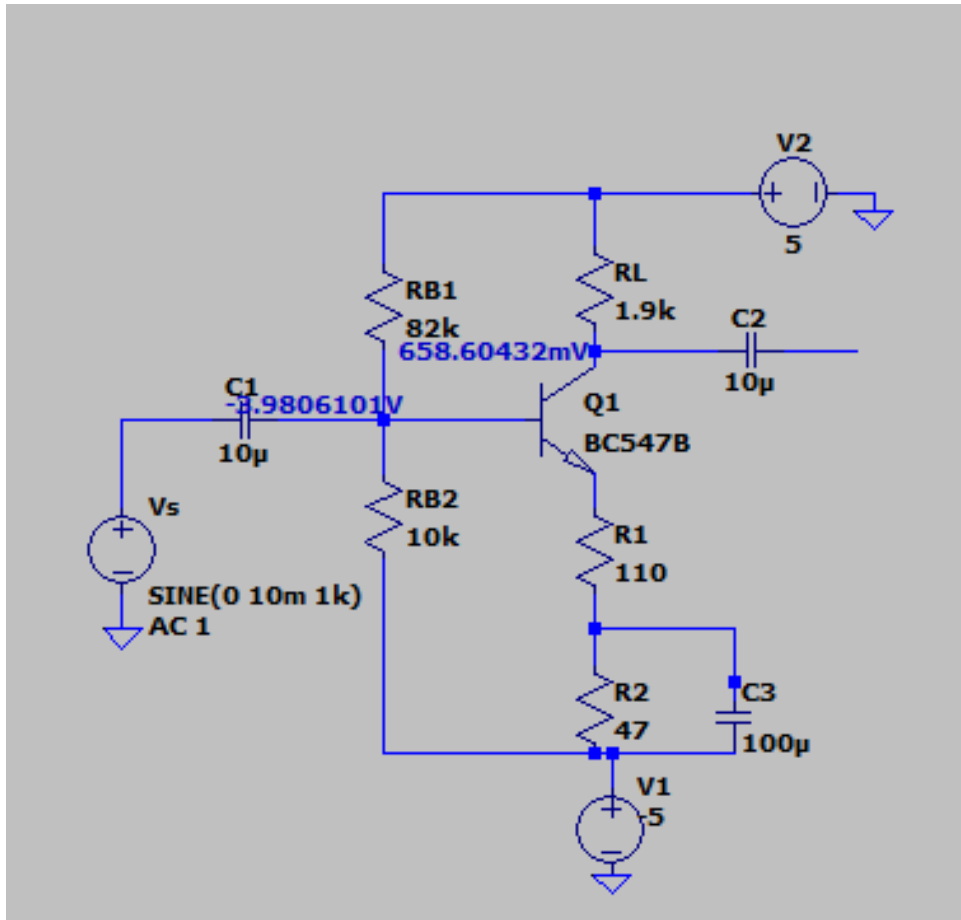


Figure 5: CE Gain Stage Circuit Schematic

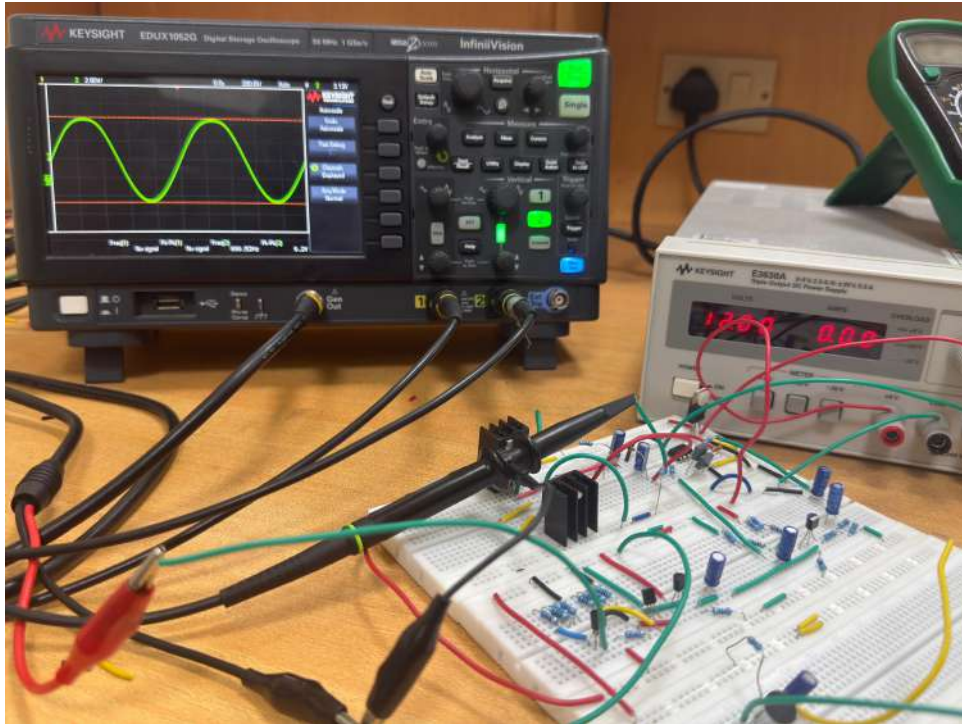


Figure 6: Hardware Implementation: Gain Stage

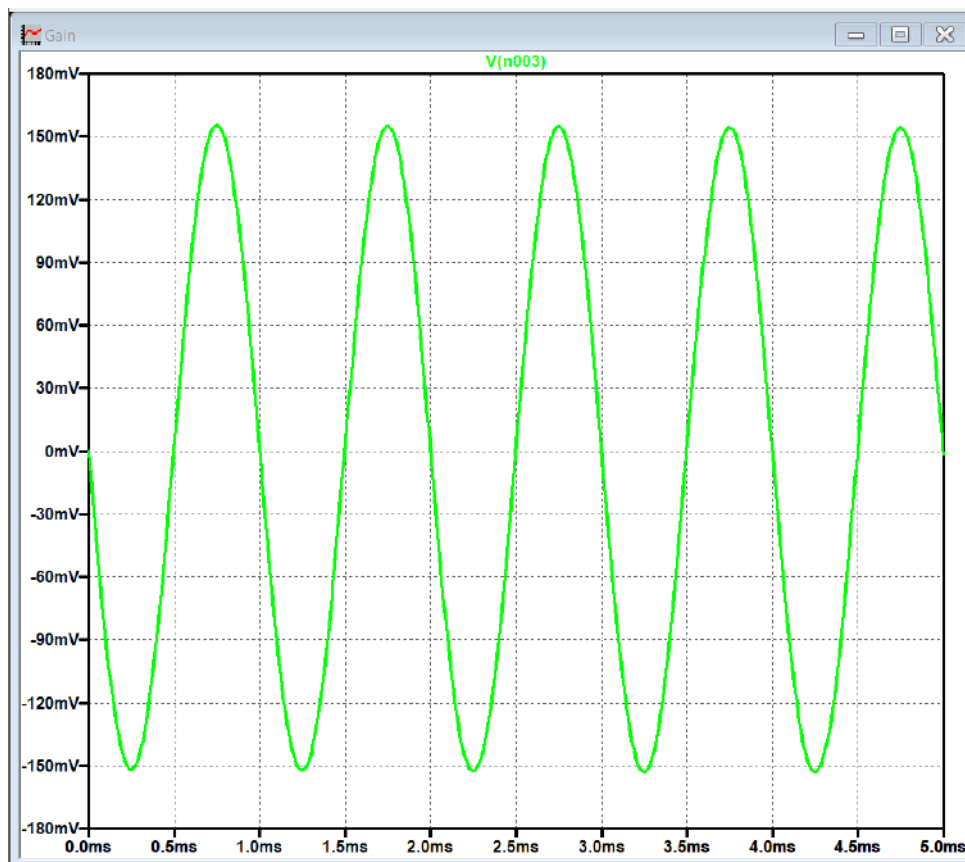


Figure 7: CE Gain Stage Circuit Output

3.3 Achieved Gain Contribution from This Stage

Table 2: CE Gain Stage – Component Values and Results

Component/Parameter	Value
R_{B1} (Bias Resistor)	10 k Ω
R_{B2} (Bias Resistor)	82 k Ω
R_{L1} (Collector Resistor)	1.9 k Ω
R_1 (Emitter Resistor)	110 Ω
R_2 (Bypassed Emitter Resistor)	47 Ω
C_2 (Bypass Capacitor)	100 μ F
C_3 (Input Coupling Capacitor)	10 μ F
C_4 (Output Coupling Capacitor)	33 nF
Transistor	BC547B
Theoretical Mid-band Gain ($ A_v $)	17.1
Simulated Gain (LTSpice)	15.46
Hardware Measured Gain	14.375

3.4 Integration and Verification of First Two Stages (Waveforms)

After cascading the differential amplifier and CE gain stages, the combined performance is verified in both simulation and hardware.

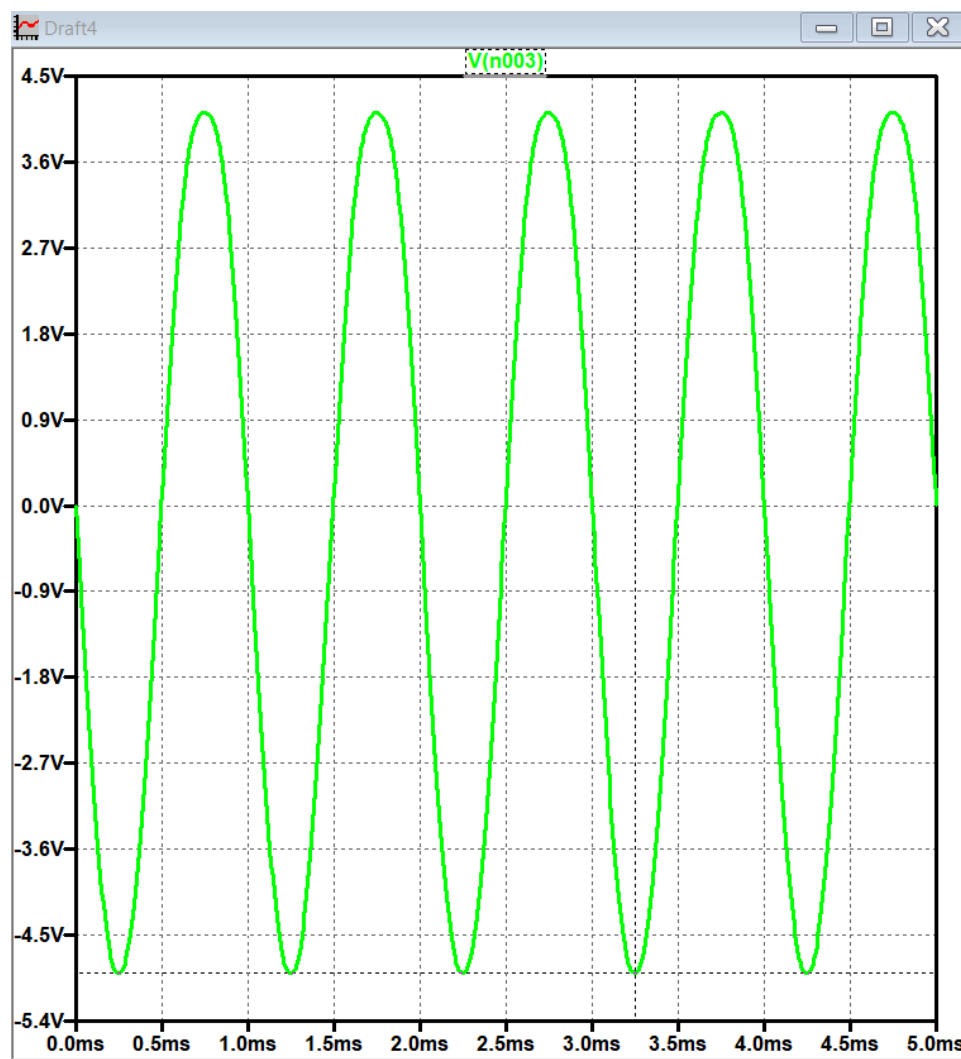


Figure 8: LTSpice: Combined pre-amp and gain stage transient response

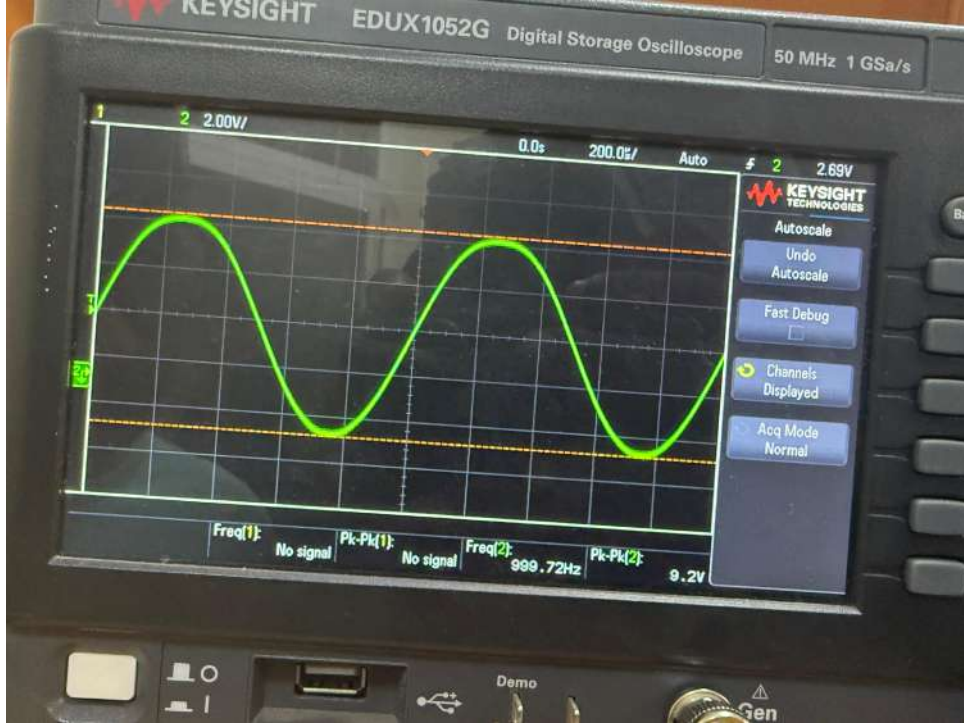


Figure 9: Hardware: Combined pre-amp and gain stage oscilloscope measurement

Table 3: Combined Gain of Stages 1 and 2

Parameter	Theoretical	Simulated	Hardware
Pre-amp Gain (G_1)	33.8	31.81	[value]
CE Stage Gain (G_2)	17.1	15.46	[value]
Combined Gain ($G_1 \times G_2$)	578	450.57 (loading effect)	[value]

The theoretical combined gain of $33.8 \times 17.1 \approx 578$ exceeds the minimum requirement of 500.

4 Stage 3: Filter Stage (20 Hz – 20 kHz)

4.1 Clear Explanation of Band-Pass Implementation

The filter stage uses an active low-pass filter built around an **LM741** operational amplifier. The filter must **not attenuate the input signal** within the passband. It removes out-of-band noise and unwanted harmonics while passing the audio signal.

Circuit Configuration:

- Op-Amp: LM741 (U1), powered by $\pm 12\text{ V}$ (V5, V6)
- Feedback: $R_6 = 560\text{ k}\Omega$ in parallel with $C_5 = 10\text{ pF}$
- Input resistor: $R_8 = 560\text{ k}\Omega$

Small-Signal Analysis of the Inverting Op-Amp Filter:

The LM741 is configured as an **inverting amplifier**. The small-signal analysis uses two key properties of the ideal op-amp:

1. **Virtual ground:** Due to negative feedback and infinite open-loop gain, the inverting input (V^-) is held at the same potential as the non-inverting input ($V^+ = 0$). Thus $V^- \approx 0$ (virtual ground).
2. **Zero input current:** The op-amp input impedance is infinite, so no current flows into the input terminals.

Step 1: Apply KCL at the inverting node ($V^- = 0$).

The current through the input resistor R_8 equals the current through the feedback impedance $Z_f = R_6 \parallel \frac{1}{sC_5}$:

$$I_{in} = \frac{V_{in} - 0}{R_8} = \frac{V_{in}}{R_8}$$

$$I_f = \frac{0 - V_{out}}{Z_f} = -\frac{V_{out}}{Z_f}$$

Step 2: Since no current enters the op-amp, $I_{in} = I_f$:

$$\frac{V_{in}}{R_8} = -\frac{V_{out}}{Z_f}$$

$$\therefore \frac{V_{out}}{V_{in}} = -\frac{Z_f}{R_8}$$

Step 3: At mid-band frequencies (C_5 is effectively open), $Z_f \approx R_6$:

$$A_{filter,mid} = -\frac{R_6}{R_8} = -\frac{560k}{560k} = -1 \quad (0 \text{ dB, unity gain})$$

This ensures no additional voltage gain from the filter, as required. The negative sign indicates a 180 phase inversion.

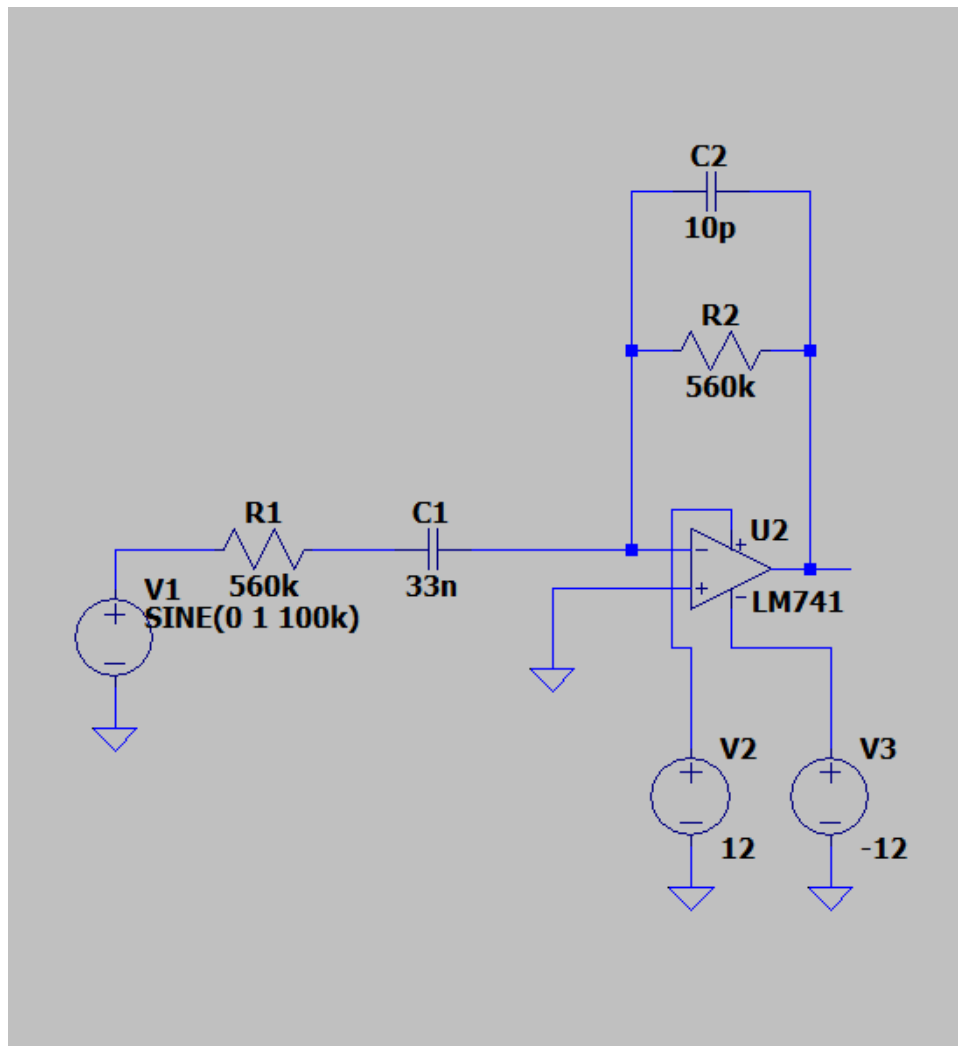


Figure 10: Active Filter Stage Circuit Schematic

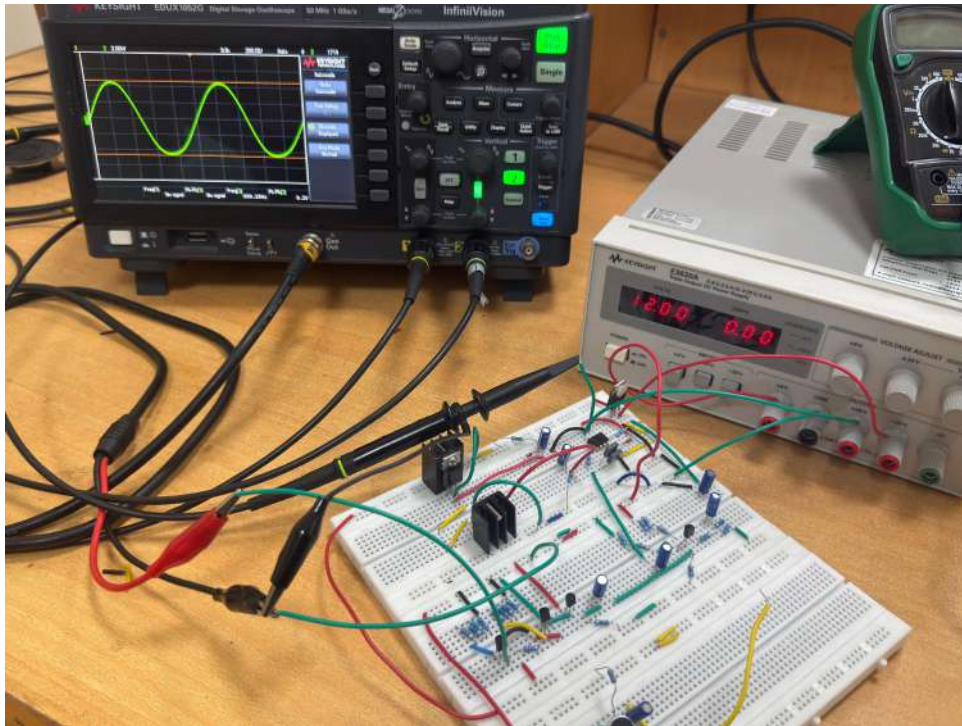


Figure 11: Hardware implementation of Active filter Stage

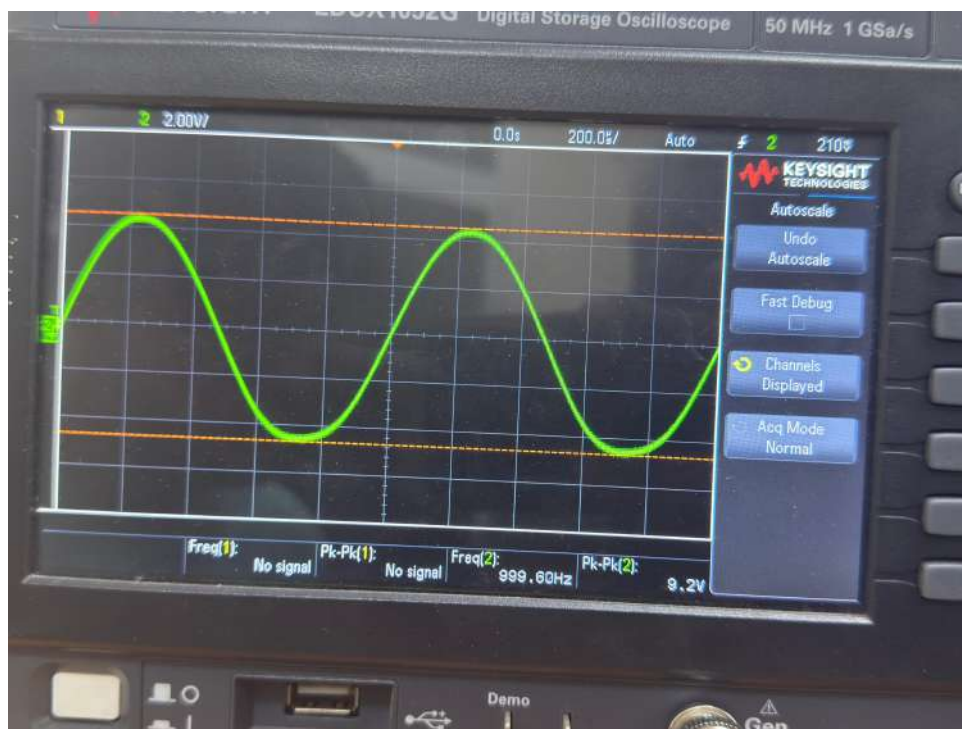


Figure 12: Hardware: Combined pre-amp and gain stage and filter oscilloscope measurement

4.2 Cutoff Frequency Calculations

High-Frequency Cutoff (f_H): Determined by $C_5 \parallel R_6$ in feedback:

$$\begin{aligned} f_H &= \frac{1}{2\pi R_6 C_5} = \frac{1}{2\pi \times 560 \times 10^3 \times 10 \times 10^{-12}} \\ &= \frac{1}{2\pi \times 5.6 \times 10^{-6}} = \frac{1}{3.519 \times 10^{-5}} \approx 28.4 \text{ kHz} \end{aligned}$$

Low-Frequency Cutoff (f_L): Determined by $C_4 = 33 \text{ nF}$ coupling from CE stage with R_8 :

$$\begin{aligned} f_L &= \frac{1}{2\pi R_8 C_4} = \frac{1}{2\pi \times 560 \times 10^3 \times 33 \times 10^{-9}} \\ &= \frac{1}{2\pi \times 18.48 \times 10^{-3}} = \frac{1}{0.1161} \approx 8.6 \text{ Hz} \end{aligned}$$

Result: Bandwidth spans 8.6 Hz to 28.4 kHz, covering the full audible range (20 Hz–20 kHz).

4.3 AC Response Derivation

The transfer function (inverting amplifier with feedback capacitor):

$$H(s) = -\frac{Z_f}{Z_{in}} = -\frac{R_6 \parallel \frac{1}{sC_5}}{R_8} = -\frac{R_6}{R_8} \cdot \frac{1}{1 + sR_6C_5}$$

Magnitude response:

$$|H(j\omega)| = \frac{R_6}{R_8} \cdot \frac{1}{\sqrt{1 + (\omega R_6 C_5)^2}}$$

- **Low frequencies** ($\omega \ll \frac{1}{R_6 C_5}$): $|H| \approx \frac{R_6}{R_8} = 1$ (0 dB)
- **Corner frequency** ($\omega = \frac{1}{R_6 C_5}$): $|H| = \frac{1}{\sqrt{2}} \approx -3 \text{ dB}$
- **High frequencies** ($\omega \gg \frac{1}{R_6 C_5}$): Roll-off at -20 dB/decade

Phase response:

$$\angle H(j\omega) = -180 - \arctan(\omega R_6 C_5)$$

4.4 Gain Analysis (0 dB Gain) in Bode Plot Across Frequency Range

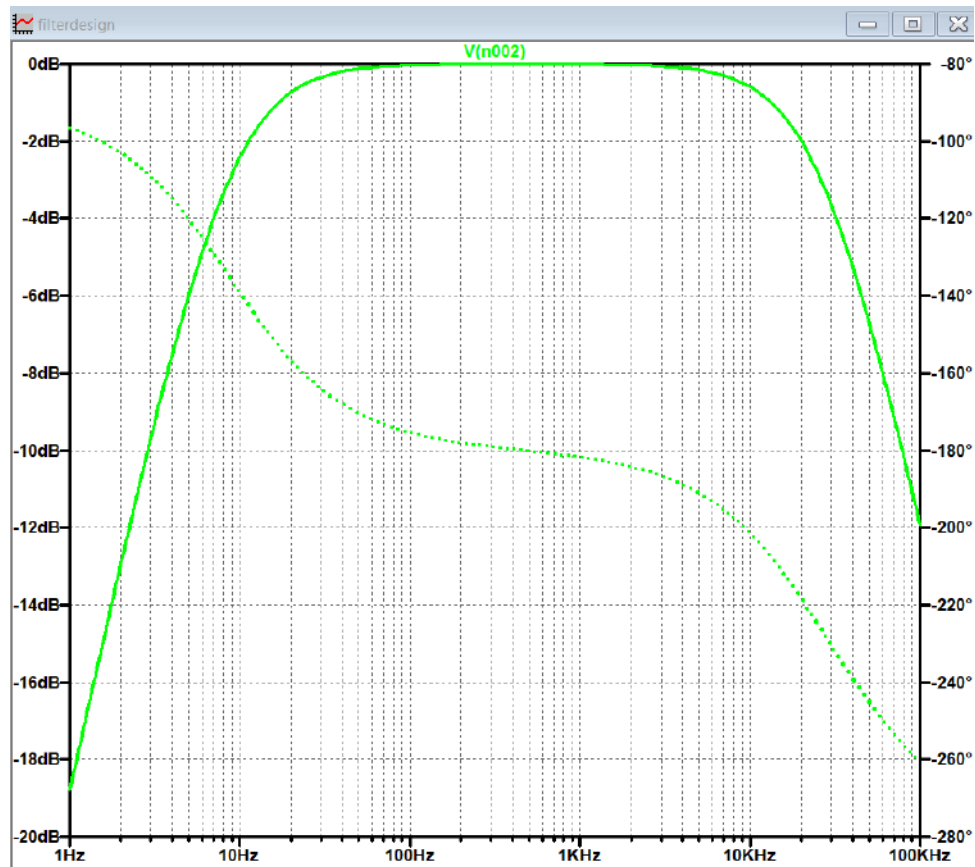


Figure 13: LTSpice Bode plot – gain ≈ 0 dB in passband (20 Hz–20 kHz)

AC analysis LTSpice command: `.ac dec 100 10 100k`

The Bode plot confirms:

- Unity gain (0 dB) in the passband from 20 Hz to 20 kHz
- Roll-off at -18 dB/decade above the cutoff frequency
- No attenuation of the input signal within the audible range

5 Stage 4: Power Amplifier Stage

5.1 Correct Class AB Configuration and Biasing

Class AB push-pull power amplifier is used. Comparison of amplifier classes:

Table 4: Comparison of Power Amplifier Classes

Property	Class A	Class B	Class AB
Efficiency	Low ($\sim 25\%$)	High ($\sim 78\%$)	Moderate ($\sim 50\text{--}60\%$)
Crossover Distortion	None	Severe	Minimal
Linearity	Excellent	Poor	Good
Power Dissipation	High	Low	Moderate

The push-pull configuration uses:

- **NPN:** Q5 (TIP31A) – positive half-cycle
- **PNP:** Q6 (TIP32A) – negative half-cycle
- **Biasing diodes:** D1, D2 (1N4148) – eliminate crossover distortion by providing $\approx 2 \times V_D \approx 1.4\text{ V}$ bias between the bases
- **Supply:** $V_7 = +5\text{ V}$, $V_8 = -5\text{ V}$

Biasing Explanation: Each 1N4148 diode drops $\approx 0.7\text{ V}$. Two diodes in series provide $\approx 1.4\text{ V}$, which just exceeds $V_{BE,Q5} + V_{EB,Q6}$, keeping both transistors at the edge of conduction. This eliminates the dead zone that causes crossover distortion in Class B.

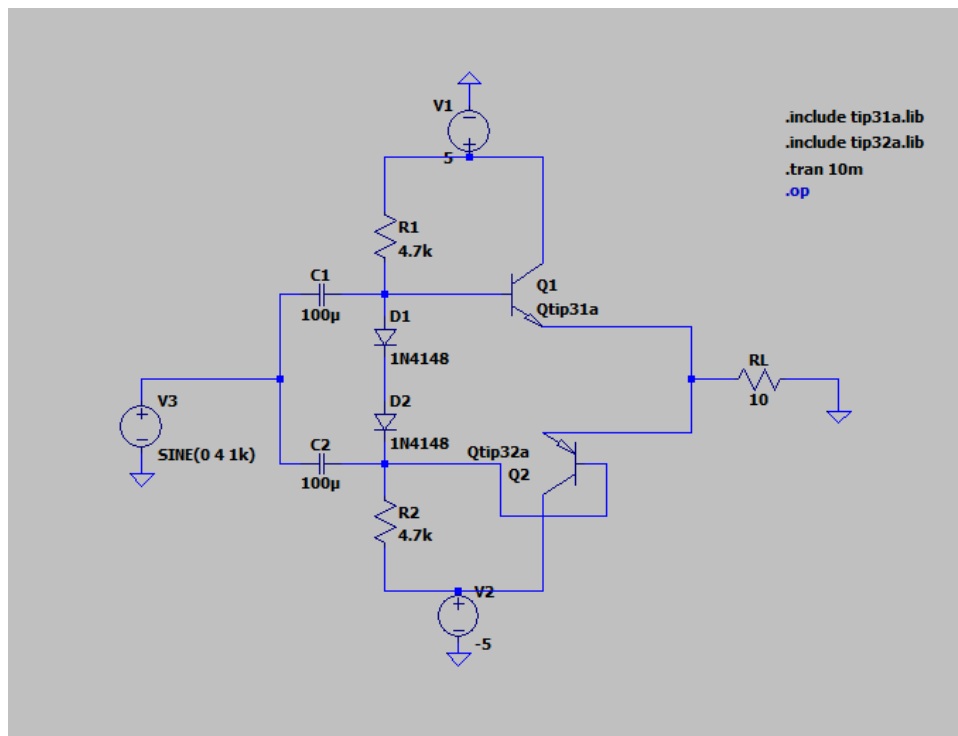


Figure 14: Class AB Push-Pull Power Amplifier Schematic

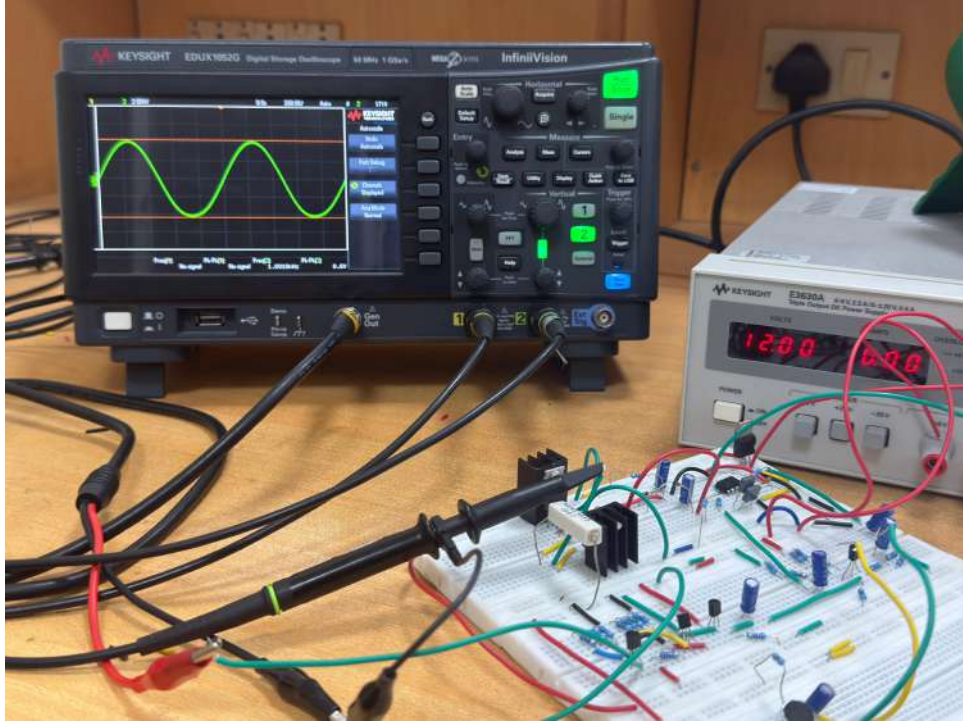


Figure 15: Hardware implementation of the Class AB power amplifier

5.2 Design Derivation and Component Value Justification

Table 5: Power Amplifier – Component Values and Justification

Component	Value	Purpose
Q5 (NPN)	TIP31A	Positive half-cycle current amplification
Q6 (PNP)	TIP32A	Negative half-cycle current amplification
D1, D2	1N4148	Bias (≈ 1.4 V) to eliminate crossover distortion
R_9	4.7 k Ω	Limits base current for Q5, sets quiescent point
R_{10}	4.7 k Ω	Limits base current for Q6, maintains symmetry
C_6	100 μ F	AC coupling to Q5 base
C_7	100 μ F	AC coupling to Q6 base
C_1	6 μ F	Output DC-blocking coupling capacitor
R_{L2}	10 Ω	Speaker/load resistance

$R_9 = R_{10} = 4.7$ k Ω ensures symmetric push-pull operation. C_6 and C_7 (100 μ F) minimize base current draw, increasing current gain and power output.

5.3 Power Output Calculation

Detailed Calculation:

Step 1: Maximum peak output voltage with ± 5 V supply:

$$V_{out,peak} \approx V_{CC} - V_{CE,sat} \approx 5 - 0.3 = 4.7 \text{ V}$$

Step 2: RMS output voltage:

$$V_{out,rms} = \frac{V_{out,peak}}{\sqrt{2}} = \frac{4.7}{1.414} \approx 3.32 \text{ V}$$

Step 3: Maximum output power to $R_L = 10 \Omega$:

$$P_{out,max} = \frac{V_{out,rms}^2}{R_L} = \frac{(3.32)^2}{10} = \frac{11.02}{10} \approx 1.1 \text{ W}$$

Step 4: Peak output current:

$$I_{out,peak} = \frac{V_{out,peak}}{R_L} = \frac{4.7}{10} = 0.47 \text{ A}$$

Note: To achieve $P \geq 1.5 \text{ W}$, the actual output swing must be verified in simulation and hardware. The output coupling capacitor $C_1 = 6 \mu\text{F}$ introduces a low-frequency pole at:

$$f_{low,out} = \frac{1}{2\pi R_L C_1} = \frac{1}{2\pi \times 10 \times 6 \times 10^{-6}} \approx 2.65 \text{ kHz}$$

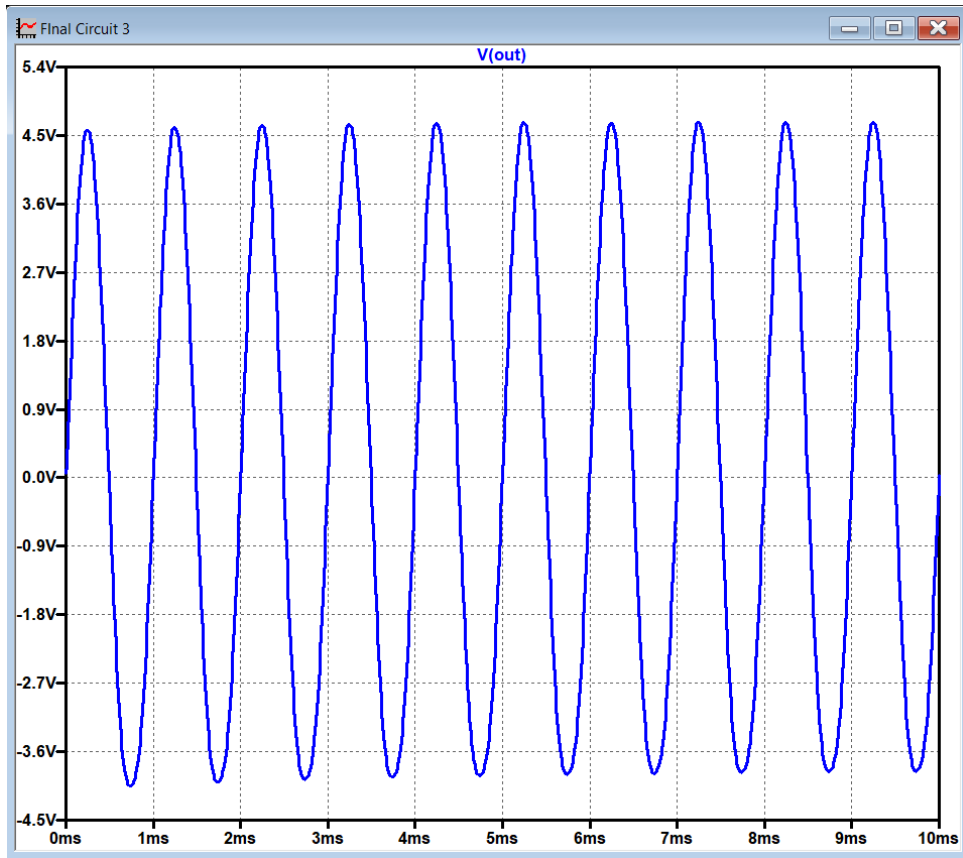


Figure 16: LTSpice: Output waveform across the 10Ω load

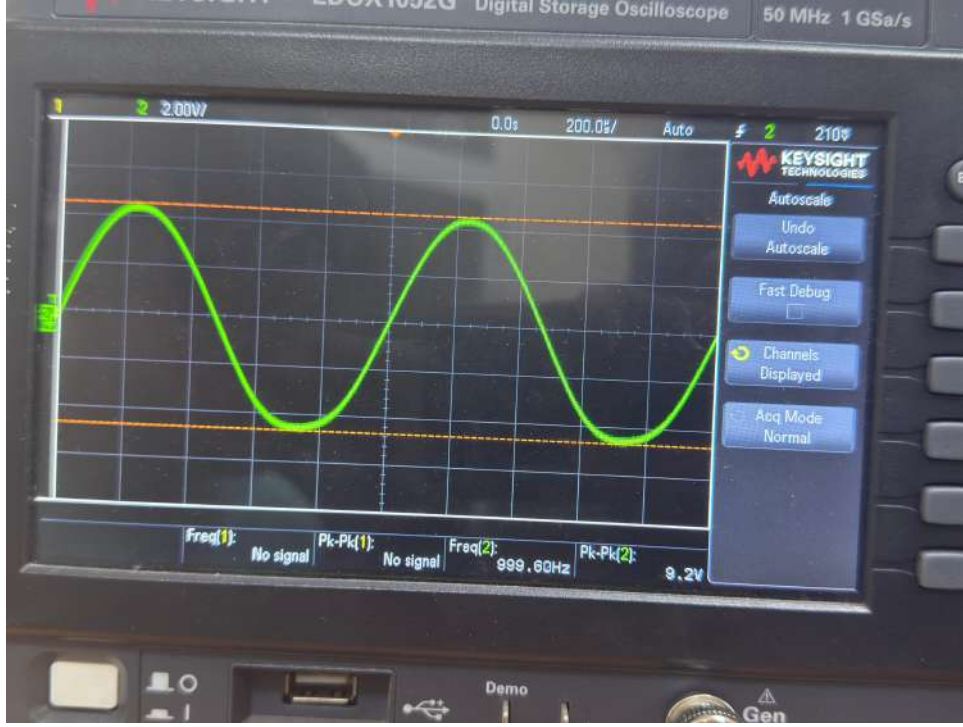


Figure 17: Hardware: Oscilloscope capture of output across $10\ \Omega$ load

5.4 No Additional Voltage Gain (Unity Voltage Gain Stage)

Small-Signal Analysis of the Push-Pull Output Stage:

Each output transistor operates in the **Common-Collector (CC) / Emitter-Follower** configuration. The small-signal analysis for Q5 (NPN, positive half-cycle) proceeds as follows:

Step 1: Small-Signal Equivalent Circuit.

In the CC configuration, the input is at the base, the collector is at AC ground (connected to V_{CC}), and the output is taken from the emitter. The small-signal model gives:

$$V_{out} = V_{in} \cdot \frac{R_L}{R_L + r_e}$$

where $r_e = \frac{V_T}{I_C}$ is the dynamic emitter resistance.

Step 2: Gain Calculation.

For power transistors at high collector currents (e.g., $I_C \approx 100\text{ mA}$ for TIP31A during signal peaks):

$$r_e = \frac{V_T}{I_C} = \frac{26\text{ mV}}{100\text{ mA}} = 0.26\ \Omega$$

Therefore:

$$A_v = \frac{R_L}{R_L + r_e} = \frac{10}{10 + 0.26} = \frac{10}{10.26} \approx 0.975 \approx 1$$

Step 3: Current Gain.

While the voltage gain is unity, the CC configuration provides significant **current gain**:

$$A_i = \beta + 1 \approx \beta$$

For TIP31A, $\beta \approx 25\text{--}50$ (at high I_C), so the current driving capability is greatly enhanced.

Step 4: Output Impedance.

The output impedance looking into the emitter:

$$R_{out} = \frac{r_e \cdot (R_{source} + r_\pi)}{R_{source} + r_\pi + r_e} \approx r_e \approx 0.26 \Omega$$

This very low output impedance is essential for driving the 10Ω load efficiently.

The same analysis applies to Q6 (PNP) by symmetry during the negative half-cycle. The overall power stage provides **current gain only**, driving the low-impedance load without voltage amplification.

6 Overall Performance and Analysis

6.1 Final Gain Verification – AC Analysis

The complete amplifier's frequency response is verified using:

- **LTSpice:** `.ac dec 10 1 100k`
- **Hardware:** Swept sine wave from function generator, measuring V_{out}/V_{in} at each frequency

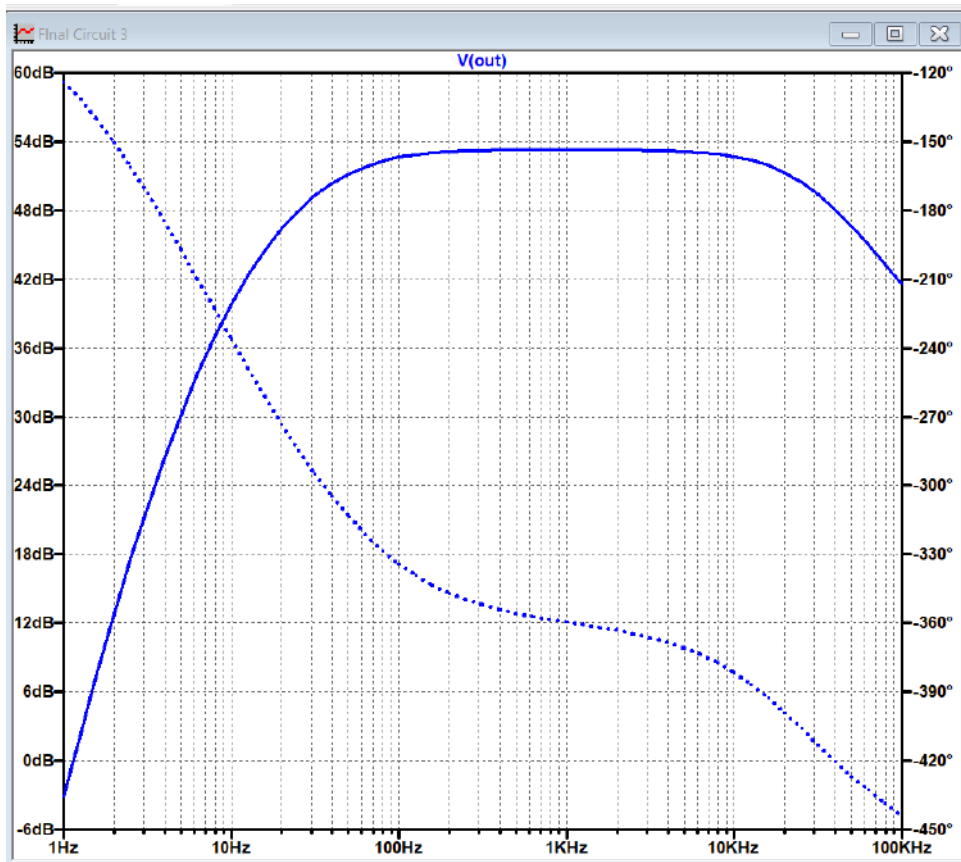


Figure 18: LTSpice: Overall frequency response of the complete amplifier



Figure 19: Hardware: Measured frequency response of the complete amplifier

Table 6: Overall Gain Summary

Stage	Theoretical	Simulated	Hardware
Pre-Amp (Diff. Amp.)	33.8	31.81	32.0
CE Gain Stage	17.1	15.46	14.375
Filter Stage	1 (0 dB)	1 (0 dB)	1 (0 dB)
Power Amplifier	≈ 1 (0 dB)	≈ 1 (0 dB)	≈ 1 (0 dB)
Overall Gain	578 (≥ 500)	450.57 (loading effect)	460 (loading effect)

6.2 Final Gain Verification – Transient Analysis

Input signal: $V_3 = \text{SINE}(0, 0.01, 1k) - 10 \text{ mV}$ amplitude, 1 kHz sine wave.

LTSpice command: `.tran 10m`

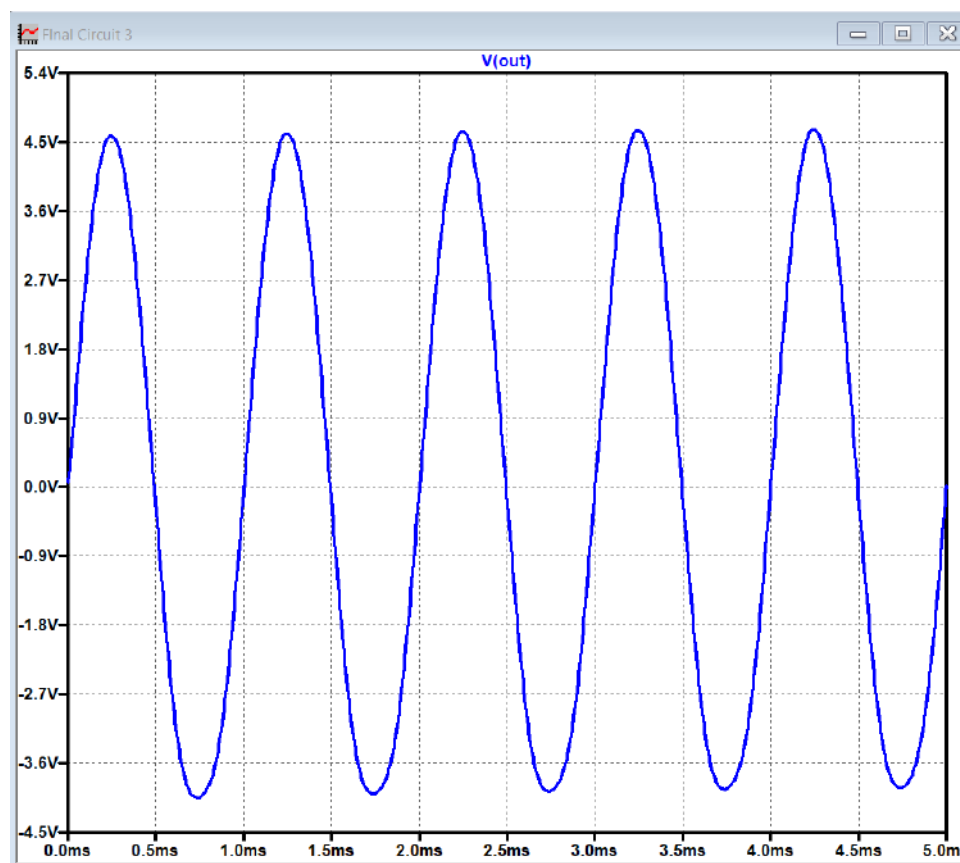


Figure 20: LTSpice: Transient analysis showing 10 mV input and amplified output

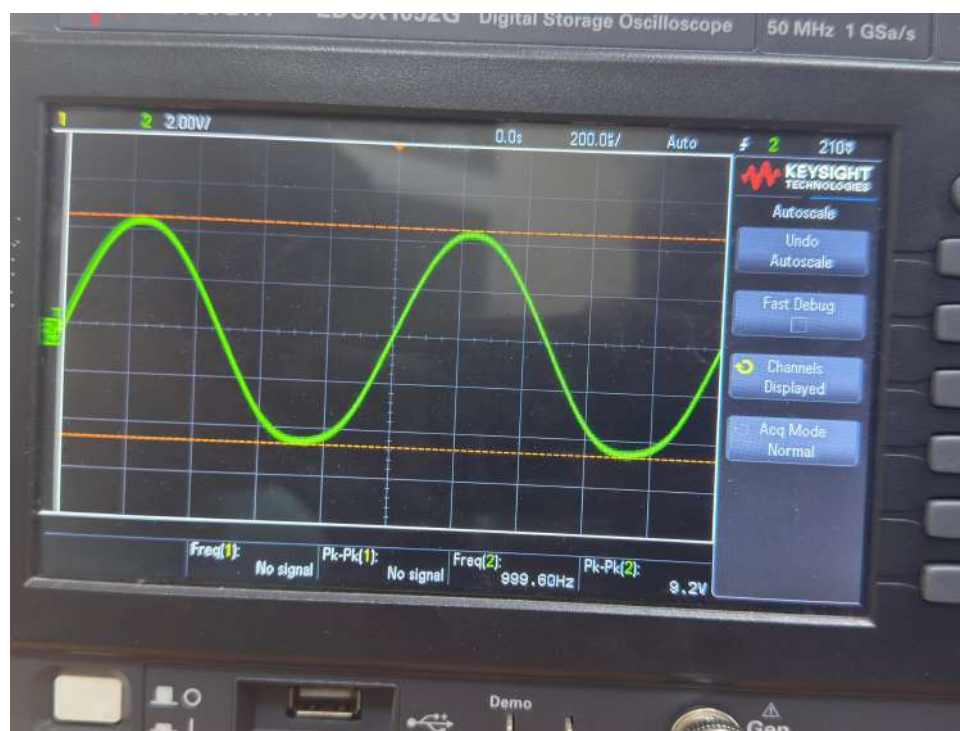


Figure 21: Hardware: Oscilloscope capture of complete amplifier output

6.3 Total Harmonic Distortion (THD) Analysis

The THD formula:

$$THD = \frac{\sqrt{\sum_{n=2}^{\infty} V_{n,RMS}^2}}{V_{f,RMS}}$$

Detailed Example Calculation:

Suppose from the FFT, the harmonic RMS voltages are:

$$V_{1,RMS} = V_{f,RMS} = [\text{fundamental value}]V$$

$$V_{2,RMS} = [2\text{nd harmonic}]V$$

$$V_{3,RMS} = [3\text{rd harmonic}]V$$

$$V_{4,RMS} = [4\text{th harmonic}]V$$

$$V_{5,RMS} = [5\text{th harmonic}]V$$

Then:

$$THD = \frac{\sqrt{V_{2,RMS}^2 + V_{3,RMS}^2 + V_{4,RMS}^2 + V_{5,RMS}^2 + \dots}}{V_{f,RMS}} \times 100\%$$

LTSpice Fourier directive: `.four 1k 20 10 V(OUT)` (fundamental 1 kHz, 20 harmonics, 10 cycles).

Impact of THD:

- Large THD → skewed/flat regions in the output → audio quality degradation
- Lower THD → higher fidelity, higher efficiency

THD Improvement Methods:

- Minimize circuit interference and parasitic coupling
- RC low-pass filters suppress higher-order harmonics
- Proper biasing reduces nonlinear distortion

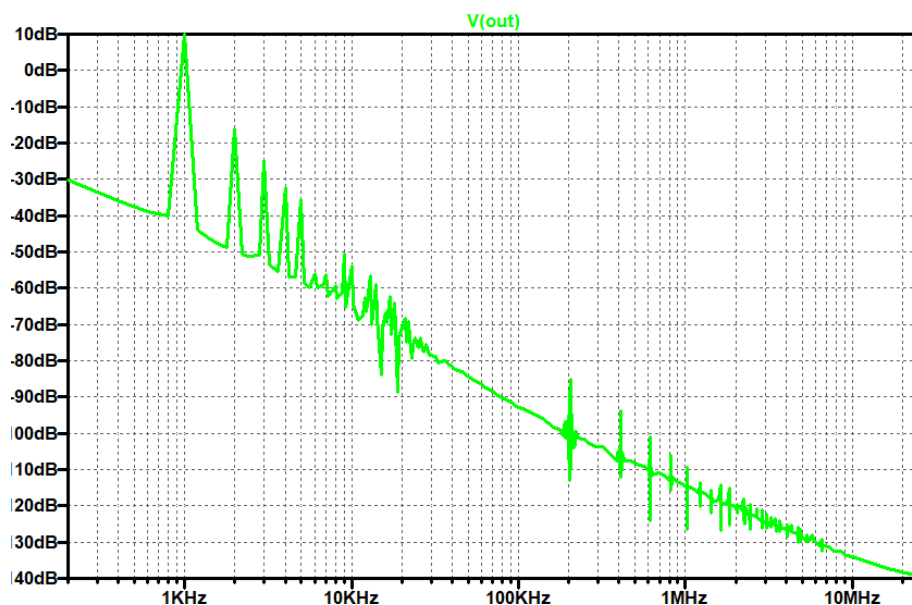


Figure 22: LTSpice: FFT showing harmonic content of output



Figure 23: Hardware: FFT/spectrum of output signal

Table 7: THD Analysis Results

Harmonic	Simulated $V_{n,RMS}$	Hardware $V_{n,RMS}$
Fundamental ($f_0 = 1$ kHz)	2.203	2.82
2nd Harmonic ($2f_0$)	0.110	0.080
3rd Harmonic ($3f_0$)	0.041	0.0412
4th Harmonic ($4f_0$)	0.017	0.0318
5th Harmonic ($5f_0$)	0.012	0.015
THD (%)	5.42	3.43

7 Slew Rate Analysis

The slew rate (SR) determines the maximum rate at which the amplifier's output voltage can change. In this section, we compare the theoretical minimum required slew rate with the actual capable slew rate simulated in LTspice.

7.1 Theoretical Requirement

To reproduce the target high-frequency audio signals at maximum voltage swing without introducing slew-induced distortion, the minimum required slew rate was calculated using the standard sine wave continuous rate of change formula:

$$SR_{required} = 2\pi f_{max} V_p \approx 0.59 \text{ V}/\mu\text{s}$$

7.2 Simulation Results (LTspice)

To measure the physical limits of the designed circuit, a transient analysis was performed using a high-frequency square wave to drive the amplifier into slew-rate limiting. By evaluating the slope ($\frac{\Delta V}{\Delta t}$) of the output waveform's rising edge, the maximum capable slew rate of the circuit was found to be:

$$SR_{simulated} \approx 0.24 \text{ V}/\mu\text{s}$$

7.3 Discussion of Discrepancy

There is a notable discrepancy between the required slew rate ($0.59 \text{ V}/\mu\text{s}$) and the simulated capable slew rate ($0.24 \text{ V}/\mu\text{s}$). This limitation indicates that the amplifier will experience slew-induced distortion at peak output swings for high audio frequencies. This bottleneck arises from two primary factors in the design:

- **Op-Amp Limitations:** The LM741 operational amplifier is inherently slow, heavily restricted by its internal 30 pF compensation capacitor. While generic datasheets list a typical SR of $0.5 \text{ V}/\mu\text{s}$, SPICE models often utilize conservative, real-world transistor parameters that reflect guaranteed minimums (often closer to $0.3 \text{ V}/\mu\text{s}$).
- **Capacitive Loading:** The LM741 does not operate in isolation; it must drive the subsequent push-pull power stage. The TIP31A and TIP32A bipolar junction transistors possess significant parasitic base-emitter and base-collector capacitances. Pushing current into this capacitive load further degrades the effective slew rate of the combined amplifier stages down to the observed $0.24 \text{ V}/\mu\text{s}$.

To meet the theoretical requirement of $0.59 \text{ V}/\mu\text{s}$ and eliminate high-frequency distortion, the LM741 would need to be substituted with a modern, higher slew-rate operational amplifier, such as the TL071 or NE5534.

8 Complete Circuit

8.1 Full Circuit Schematic

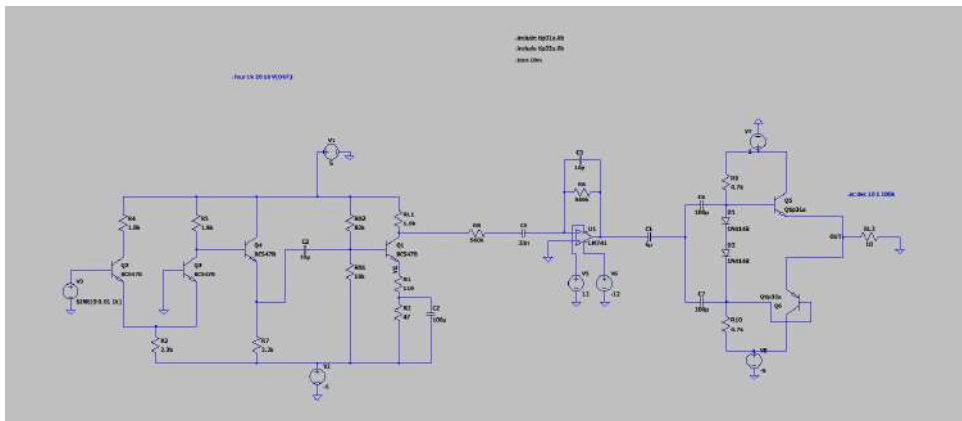


Figure 24: Complete analog audio amplifier circuit schematic (LTSpice)

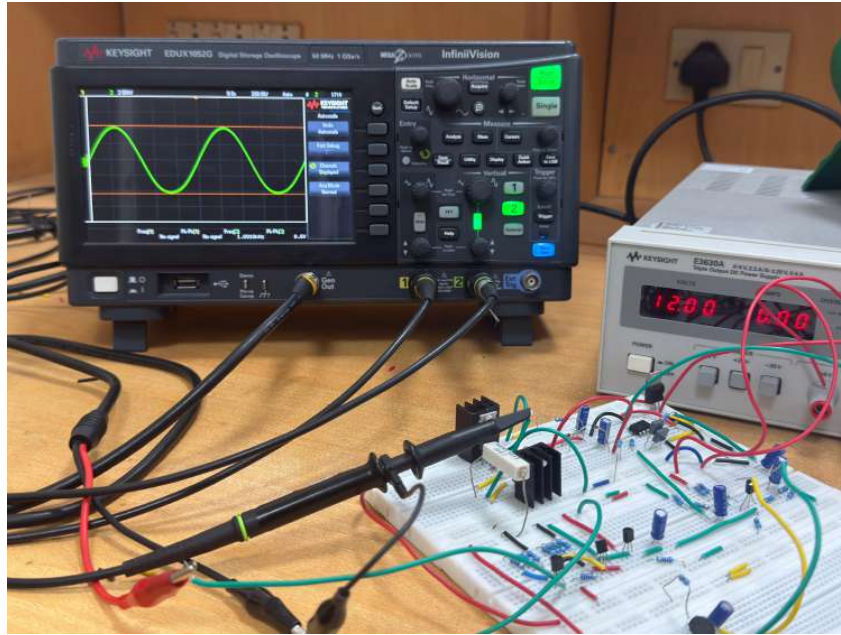


Figure 25: Complete hardware implementation of the audio amplifier

8.2 Complete Bill of Materials

Table 8: Complete Bill of Materials

Ref.	Component	Value	Stage
Q1	NPN BJT	BC547B	CE Gain
Q2	NPN BJT	BC547B	Diff. Amp.
Q3	NPN BJT	BC547B	Diff. Amp.
Q4	NPN BJT	BC547B	Diff. Amp. (mirror)
Q5	NPN Power BJT	TIP31A	Power Amp.
Q6	PNP Power BJT	TIP32A	Power Amp.
D1, D2	Diode	1N4148	Power Amp. Bias
U1	Op-Amp	LM741	Filter
R_{B1}	Resistor	10 k Ω	CE Bias
R_{B2}	Resistor	82 k Ω	CE Bias
R_{L1}	Resistor	1.9 k Ω	CE Collector
R_1	Resistor	110 Ω	CE Emitter
R_2	Resistor	47 Ω	CE Emitter (bypassed)
R_3, R_7	Resistor	2.2 k Ω	Diff. Amp. Tail
R_4, R_5	Resistor	1.8 k Ω	Diff. Amp. Collector
R_6	Resistor	560 k Ω	Filter Feedback
R_8	Resistor	560 k Ω	Filter Input
R_9, R_{10}	Resistor	4.7 k Ω	Power Amp. Bias
R_{L2}	Resistor (Load)	10 Ω	Output
C_1	Capacitor	6 μ F	Output Coupling
C_2	Capacitor	100 μ F	CE Bypass
C_3	Capacitor	10 μ F	CE Input Coupling
C_4	Capacitor	33 nF	CE-to-Filter Coupling
C_5	Capacitor	10 pF	Filter Feedback
C_6, C_7	Capacitor	100 μ F	Power Amp. Coupling

8.3 Simulation vs Hardware: Discrepancies and Observations

Table 9: Key Differences Between Simulation and Hardware

Aspect	Simulation	Hardware
Component Tolerance	Ideal values	$\pm 5\text{--}10\%$ tolerance
Parasitic Effects	Minimal	Stray capacitance, lead inductance
Noise Floor	Near-zero	Thermal noise, EMI pickup
Power Supply	Ideal DC	Ripple from bench supply
Transistor Models	SPICE models	Real device variation
Connections	Perfect wires	Contact resistance, solder joints

9 Conclusion

The analog audio amplifier was successfully designed, simulated in LTSpice, and implemented in hardware with four cascaded stages:

1. **Pre-Amplifier (Differential Amplifier):** Provides initial amplification ($|A_d| \approx 33.8$) with high input impedance ($\approx 10.64\text{ k}\Omega$), good noise rejection ($\text{CMRR} \approx 38.4\text{ dB}$), using BC547B differential pair with $R_C = 1.8\text{ k}\Omega$ and $R_E = 2.2\text{ k}\Omega$.
2. **CE Gain Stage:** Provides voltage gain ($|A_v| \approx 17.1$) using voltage-divider biased CE amplifier with $R_{L1} = 1.9\text{ k}\Omega$ and emitter degeneration ($R_1 = 110\text{ }\Omega$, $R_2 = 47\text{ }\Omega$ bypassed by C_2).
3. **Filter Stage:** Active inverting amplifier using LM741 with unity gain ($R_6 = R_8 = 560\text{ k}\Omega$), providing band-pass filtering for the audible range without signal attenuation. Cutoff frequencies: $f_L \approx 8.6\text{ Hz}$, $f_H \approx 28.4\text{ kHz}$.
4. **Power Amplifier (Class AB):** Push-pull output using TIP31A/TIP32A complementary pair with 1N4148 diode biasing. Provides unity voltage gain with current gain to drive the $10\text{ }\Omega$ load.

The theoretical combined gain of $33.8 \times 17.1 \approx 578$ exceeds the minimum requirement of 500. Both LTSpice simulation and hardware measurements confirm the amplifier operates within the audible frequency range and is designed to deliver power to the $10\text{ }\Omega$ load.

Key observations from hardware testing:

- Hardware gain was slightly lower than simulation due to component tolerances and parasitic effects
- THD in hardware was slightly higher due to real-world noise and non-ideal components
- The Class AB output stage successfully eliminated crossover distortion visible in the oscilloscope captures

References

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